

Yash Rupani

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SUMMARY

Master's graduate with 2+ years of enterprise experience building scalable **ETL pipelines** and **real-time data architectures** on **AWS, GCP, and Microsoft Fabric**. Expert in **Apache Spark, Kafka, and AI-driven data systems** (RAG, Vector DBs). Proven track record of reducing infrastructure costs and optimizing data retrieval latency.

PUBLICATIONS & CERTIFICATIONS

Publication: Co-author, "Computational Alliance Modeling for Conversational AI" (Accepted at AMCIS 2026)

Certifications: Microsoft Certified: Fabric Data Engineer Associate | Snowflake Data Warehousing | Databricks AI Agent

TECHNICAL SKILLS

Languages: Python, SQL, Bash, C++

Data Engineering: Apache Spark, PySpark, Kafka, Airflow, Snowflake, dbt, Microsoft Fabric, Pandas, NumPy

Cloud & DevOps: AWS (S3, Lambda, Glue, EMR, Redshift), GCP (BigQuery, Firestore), Azure (Microsoft Fabric), Docker

AI & Databases: Vector Databases (FAISS), RAG Architectures, AI Agents, PostgreSQL, NoSQL, Dimensional Modeling

PROFESSIONAL EXPERIENCE

Software Engineering Intern

May 2026 – Present

QuickGrants

Remote, USA

- Engineered an **automated rule generation pipeline** using **OCR and text chunking**, processing **100+ page policy documents** and reducing manual data extraction time by **80%**.
- Architected **backend parsing workflows** to transform unstructured text into **structured reports**, accelerating the preparation time for technical product demonstrations by **15 hours per week**.
- Optimized the **data extraction accuracy** of the parsing tool to **95%**, ensuring reliable and formatted outputs for **client-facing engineering prototypes**.

Research Assistant – Data Engineering Focus

May 2026 – Present

Oregon State University

OR, USA

- Engineered an end-to-end autonomous ETL pipeline for unstructured multimedia data, leveraging **Python-based preprocessing** to eliminate **40%** of manual data preparation for **Agentic AI workflows**.
- Refined backend data structures and indexing strategies for **AI Agents**, slashing **retrieval latency by 25%** and boosting **real-time inference performance**.
- Developed robust **data validation frameworks** to ensure **100% data integrity** while enabling scalable ingestion of **terabyte-scale datasets**.

Sr. Systems Engineer (Data Engineering)

Jun. 2021 – Jun. 2023

Infosys (Client: Charter Communications)

Maharashtra, India

- Spearheaded the modernization of enterprise **data quality pipelines**, automating workflows to eliminate **10+ hours/week** of manual intervention.
- Designed high-performance **ETL workflows** using **Apache Spark** to process large-scale system monitoring data, compressing **runtime by 40%** per cycle.
- Constructed dynamic dashboards from semi-structured logs, replacing manual analysis with **automated monitoring solutions**.

PROJECTS

Shop Pulse: Real-Time E-Commerce Lakehouse (Spark, Kafka, Python, Docker) [\[Live Demo\]](#)

Jan. 2026 – Mar. 2026

- Architected a **Lakehouse pipeline** leveraging **Spark Streaming** and **Kafka**, achieving **sub-second latency** for live data visualization.
- Deployed a **fault-tolerant storage layer** using partitioned **Parquet** files and **Docker containerization**, optimizing **query performance** for downstream analytics.

Crypto Sentinel: Real-Time Market Monitor (Next.js, FastAPI, Spark, Kafka, MongoDB) [\[Live Demo\]](#)

Feb. 2026

- Engineered a **real-time streaming platform** using **FastAPI** and **Kafka** to ingest market data, reducing **data latency by 40%** compared to traditional polling methods.
- Built a robust **WebSocket alert system** processing **500+ events/second**, triggering instant notifications when asset prices breach user-defined thresholds.

EDUCATION

Master of Engineering in Computer Science, Oregon State University, USA

Sep. 2023 – Dec. 2025

Bachelor of Technology in Electrical Engineering, PDEU, India

Aug. 2017 – Jun. 2021